324652(24)

CSVTUonline.com

B. E. (Sixth Semester) Examination, Nov.-Dec. 2015

(New Scheme)

(Elect. Engg. Branch)

ELECTRICAL MACHINES-III

Time Allowed : Three hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : *Attempt all questions. Part (a) from each question is compulsory. Attempt any two parts from parts (b), (c) and (d) of each question.*

CSVTUonline.com

1. (a) Give basic reason of using transformation in electrical machines. 2

(b) Derive park two axis transformation. 7

(c) Show how a tree phase winding with variable inductance of 3ϕ, 2 pole machine can be converted to machine with winding having constant inductances. 7

(d) Derive the transient power angle equation of salient pole synchronous machine. 7

CSVTUonline.com

2. (a) Explain power invariance as applied to electrical machine analysis. 2

(b) For three phase induction motor derive : 7

$$V_{1d} = r_1 z_{1d} + P\lambda_{1q} - \lambda_{1q} P\theta$$

$$V_{1q} = r_1 z_{1q} + P\lambda_{1q} + \lambda_{1d} P\theta$$

$$V_{2d} = r_2 i_{2d} + P\lambda_{2q} - \lambda_{2q} P\theta_s \quad \text{where} \quad P\theta = W$$

$$V_{2q} = r_2 i_{2q} + \lambda_{2q} + \lambda_{2d} P\theta_s \quad P\theta_s = SW$$

(c) From steady state equivalent circuit of 3-phase induction motor derive an expression for maximum torque. 7

CSVTUonline.com

(d) Derive equivalent circuit of three phase induction motor to stimulate single phasing from stator side. 7

3. (a) What is fractional h.p. motors? 2

CSVTUonline.com

(b) A single-phase, 4-pole induction motor takes a line current of $60\angle -70A$ at standstill with its main winding excited from 230 V, 50 Hz source. Neglecting stator impedance, magnetizing current and rotational loss. Compute the torque at a slip of 0·05. 7

(c) Describe the methods to make single phase induction motor self staring. 7

(d) Develop the equivalent circuit of single phase induction motor based on double revolving field theory. 7

4. (a) What is universal motor and why it is used only in fractional horse power range. 2

CSVTUonline.com

(b) Explain how the performance analysis of two phase induction motor can be made when the supply is unbalance with the help of two phase symmetrical component. 7

(c) A 5 h.p. universal motor has $r = 30\,\Omega$ and total inductance of 0·5 H. When connected to 250 V d.c. supply and loaded to take 0·8 A, its speed is 2000 r.p.m. Find the speed and power factor when

connected to 250 V, 50 Hz a.c. supply and loaded to take same current. 7

CSVTUonline.com

(d) Explain the process of commutation a.c. series motor. 7

5. (a) How can the speed of linear induction motor controlled. 2

(b) Explain in brief stepper motor. 7

(c) Describe three phase, three pulse brushless d.c. motor. 7

(d) Explain construction and working principles of reluctance motor. 7

CSVTUonline.com